

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>

Number of vacuum cleaners sold at the green house = 1 / 3 of the total

Let's say Melanie started with x vacuum cleaners .

At the green house , she sold x / 3 vacuum cleaners .

Remaining after the green house = $x - x / 3 = 2x / 3$

At the red house , she sold 2 more vacuum cleaners , so she has $2x / 3 - 2$ vacuum cleaners left .

At the orange house , she sold half of what was left , which is $(2x / 3 - 2) / 2 = x / 3 - 1$ vacuum cleaners .

She has 5 vacuum cleaners left , so the total number of vacuum cleaners she sold is $x - 5$.

The total number of vacuum cleaners sold is the sum of the vacuum cleaners sold at the green house , red house , and orange house :

$x / 3 + 2 + (x / 3 - 1) = x / 3 + x / 3 + 1 = 2x / 3 + 1$

We know that the total number of vacuum cleaners sold is equal to the number of vacuum cleaners she started with minus the number of vacuum cleaners she has left :

$2x / 3 + 1 = x - 5$

To solve for x , we can multiply both sides by 3 to eliminate the fraction :

$6x + 3 = 3x - 15$

Subtracting 3x from both sides gives us :

$3x + 3 = -15$

Subtracting 3 from both sides gives us :

$3x = -18$

Dividing both sides by 3 gives us :

$x = -6$

However , the number of vacuum cleaners can't be negative , so this is not a valid solution .

Let's try a different approach :

The number of vacuum cleaners sold at the green house is 1 / 3 of the total , so the number of vacuum cleaners left is 2 / 3 of the total .

The number of vacuum cleaners sold at the red house is 2 , so the number of vacuum cleaners left is 2 / 3 of the total minus 2 .

The number of vacuum cleaners sold at the orange house is 1 / 2 of what's left , so the number of vacuum cleaners left is 1 / 2 of 2 / 3 of the total minus 1 .

We know that the number of vacuum cleaners left is 5 , so we can set up the equation :

$2 / 3 \text{ of the total } - 2 = 1 / 2 \text{ of } 2 / 3 \text{ of the total } - 1$

Multiplying both sides by 6 to eliminate the fractions :

$4 \text{ of the total } - 12 = 3 \text{ of the total } - 3$

Adding 12 to both sides gives us :

$4 \text{ of the total } = 9$

Subtracting 9 from both sides gives us :

$-5 \text{ of the total } = 0$

Multiplying both sides by -1 gives us :

$5 \text{ of the total } = 0$

Adding 5 to both sides gives us :

$0 \text{ of the total } = 5$

This is not a valid equation , so we must have made a mistake somewhere .

Let's try a different approach :

Let's say Melanie started with x vacuum cleaners .

At the green house , she sold x / 3 vacuum cleaners .

At the red house , she sold 2 vacuum cleaners .

At the orange house , she sold $(2x / 3 - 2) / 2$ vacuum cleaners .

The total number of vacuum cleaners she sold is $x / 3 + 2 + (2x / 3 - 2) / 2 = x / 3 + 2 + x / 3 - 1 = 2x / 3 + 1$.

The total number of vacuum cleaners she has left is 5 , so the total number of vacuum cleaners she sold is $x - 5$.

We know that the total number of vacuum cleaners she sold is equal to $2x / 3 + 1$, so we can set up the equation :

$x - 5 = 2x / 3 +$